# 👗 Fashion Sales Data Analysis

 <img src="https://img.shields.io/badge/Python-3.x-blue?style=for-the-badge&logo=python"/> <img src="https://img.shields.io/badge/Jupyter-Notebook-orange?style=for-the-badge&logo=jupyter"/> <img src="https://img.shields.io/badge/Status-Completed-brightgreen?style=for-the-badge"/> 

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## 📌 Project Overview

This project analyzes dress attributes and predicts whether a dress will be recommended based on sales data. The dataset contains attributes of dresses such as style, price, rating, season, material, and fabric type, along with customer recommendation data.

Sales are monitored on the basis of alternate days. The project focuses on **data cleaning**, **exploratory data analysis (EDA)**, and **visualization** to uncover patterns in customer preferences, pricing trends, and product recommendations.

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## 📁 Dataset Information

| Detail | Info |
|---|---|
| **Source Format** | ARFF (Attribute-Relation File Format) |
| **Tool Origin** | Weka Machine Learning Tool |
| **Total Columns** | 13 (after renaming) |
| **Target Variable** | Recommendation (1 = Recommended, 2 = Not Recommended) |

### 📋 Columns

`Style`, `Price`, `Rating`, `Size`, `Season`, `NeckLine`, `SleeveLength`, `Waistline`, `Material`, `FabricType`, `Decoration`, `PatternType`, `Recommendation`

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## 🛠️ Tools & Libraries Used

| Library | Usage |
|---|---|
| `scipy.io (arff)` | Loading the raw ARFF dataset |
| `NumPy` | Numerical operations, NaN handling |

| Library | Usage |
| --- | --- |
| `Pandas` | Data cleaning, manipulation, aggregation |
| `Matplotlib` | Static visualizations (bar, histogram, scatter) |
| `Seaborn` | Statistical plots with KDE |
| `Plotly` | Interactive multivariate charts & heatmaps |

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## 🔄 Project Workflow

### ✅ Step 1: Data Loading

- Loaded dataset from ARFF format using `scipy.io.arff`
- Converted to Pandas DataFrame
- Saved a copy as CSV for reference

### ✅ Step 2: Data Cleaning

- Decoded byte strings to UTF-8
- Renamed columns to meaningful names
- Replaced `?` placeholders with `NaN`
- Filled missing **text columns** with **Mode**
- Replaced **zero ratings** (120 found) with **Median**
- Fixed text casing and whitespace inconsistencies
- Corrected spelling errors in `Season`, `SleeveLength`, `Material`, `FabricType`
- Removed duplicate rows
- Corrected data types for all columns

### ✅ Step 3: Feature Engineering

Created 3 new derived columns:

- **RatingCategory** — Poor / Good / Very Good / Excellent
- **PriceLevel** — Numeric mapping of Price (1–5)
- **IsRecommended** — Text label from Recommendation column

### ✅ Step 4: Filtering & Aggregation

- Filtered Recommended, Not Recommended, High Rated, Casual, and Summer dresses

- Aggregated average ratings by Style and Season
- Counted dress distribution by Style

### ✅ Step 5: Exploratory Data Analysis (EDA)

#### 📊 Univariate Analysis

- Rating distribution histogram
- Recommendation, Style, Price, Season, Size bar charts

#### 📊 Bivariate Analysis

- Rating vs Recommendation
- Rating vs Season (line chart)
- Rating vs Price (scatter chart)
- Season vs Recommendation
- Style vs Recommendation

#### 📊 Multivariate Analysis *(Interactive — Plotly)*

- Style + Season vs Recommendation
- Price + Season vs Average Rating
- Style + Price vs Recommendation (heatmap)

### ✅ Step 6: Correlation Analysis

- Computed correlation matrix for `Rating`, `Recommendation`, `Price_num`
- Visualized using an interactive Plotly heatmap

### ✅ Step 7: Outlier Treatment

- Used IQR method to detect outliers in the Rating column
- Applied threshold-based removal (Rating < 3.5)
- Verified dataset after removal

### ✅ Step 8: Final Visualization

- Plotted Rating Distribution after outlier removal using Seaborn histplot with KDE

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## 📈 Key Insights

- 📌 Ratings are **heavily concentrated between 4.5 – 4.8**, showing overall positive customer

reception

- 📌 **Casual style** is the most frequently occurring dress style
- 📌 **Higher price does not guarantee higher ratings** — style and quality matter more
- 📌 **Seasonal combinations** significantly influence recommendation rates
- 📌 Rating and Recommendation show a **weak-to-moderate positive correlation**
- 📌 After outlier removal, the dataset became cleaner and more consistent for analysis

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## ✅ Conclusion

This project successfully performed end-to-end data cleaning, feature engineering, and exploratory data analysis on a Fashion Sales dataset. Starting from a raw ARFF file with noisy, inconsistent data, the dataset was transformed into a clean, analysis-ready format.

The EDA revealed that customer ratings are concentrated in the high range (4.5–4.8), suggesting overall positive reception of the fashion catalog. However, recommendation is not solely driven by rating or price — factors like style preference, season, and fabric play equally important roles.

This analysis provides a strong foundation for building a **recommendation prediction model** or guiding **inventory and design decisions** based on actual customer preferences.

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## 📂 Project Structure

```
📁 Fashion-Sales-Data-Analysis/
├── Fashion_Sales_data.ipynb        # Main Jupyter Notebook
├── fashions sales.csv              # Raw dataset (CSV copy)
├── fashion_sales_cleaned.csv       # Final cleaned dataset
└── README.md                       # Project documentation
```

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## 👩‍💻 Author

> Final Year Data Analytics Assignment
> Tools: Python | Pandas | Seaborn | Matplotlib | Plotly | Jupyter Notebook